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Engineering virus-like nanocarriers for biomolecule delivery

Abstract

Disclosed herein are modified capsid proteins comprising a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein. Disclosed are capsids comprising a plurality of modified capsid proteins. Disclosed are multimeric assemblies comprising a plurality of capsids within a membrane. Also disclosed are modified non-enveloped viruses comprising a capsid wherein the capsid comprises a plurality of modified capsid proteins, wherein the plurality of modified capsid proteins comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the modified capsid protein, wherein the capsid forming protein is a capsid forming protein of a non-enveloped virus.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of U.S. Application No. 62/751,457, filed on Oct. 26, 2018. The content of this earlier filed application is hereby incorporated by reference herein in its entirety.

REFERENCE TO SEQUENCE LISTING

(1) The Sequence Listing submitted Jan. 22, 2020 as a text file named “21101_0379U2_Sequence_Listing.txt,” created on Jan. 10, 2020, and having a size of 138,564 bytes is hereby incorporated by reference pursuant to 37 C.F.R. § 1.52(e)(5).

BACKGROUND

(2) Viruses have traditionally been categorized as either “enveloped” or “non-enveloped”, but in recent years this distinction has become blurred with the identification of “quasi-enveloped” viruses. In the conventional view, enveloped viruses contain single capsids surrounded by a membrane that they acquire, together with transmembrane envelope glycoproteins, as they bud from producer cells (FIG. 1A). These virus-encoded envelope glycoproteins are important for infection because they bind receptors and mediate the membrane fusion reactions for capsids to enter target cells (FIG. 1A). Envelope glycoproteins are also typically the major targets for neutralizing antibodies. In contrast, non-enveloped viruses were traditionally thought to exit producer cells only upon cell lysis (FIG. 1B). Unlike enveloped viruses, their capsids typically contain all of the elements required to bind receptors and enter target cells by disrupting their membrane (rather than via membrane fusion reactions). Recently, however, a subset of the viruses that were traditionally considered to be “non-enveloped” were found to be released from cells within membrane vesicles, often with multiple capsids within each vesicle (FIG. 1C). This newly described class of viruses are now referred to as “quasi-enveloped” viruses (FIG. 1C). Unlike classic enveloped viruses, quasi-enveloped virions retain the ability to enter cells through membrane disruption mechanisms (rather than membrane fusion), yet they also remain fully infectious when packaged inside vesicles. Current models hold that quasi-enveloped viruses are endocytosed by target cells, and their infectious internal capsids are then released when the vesicle begins to break down within the endosome. Once freed, the “naked” capsids retain their intrinsic ability to cross the endosomal membrane, and can thereby enter the cytoplasm and initiate infections. Importantly, quasi-enveloped viruses appear to have evolved advantages associated with both enveloped and non-enveloped viruses because they do not need viral fusion proteins to enter cells, yet their membrane coats protect the internal capsids from antibody recognition.

BRIEF SUMMARY

(3) Disclosed herein are modified capsid proteins comprising a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein.

(4) Disclosed herein are capsids comprising a plurality of the disclosed modified capsid proteins.

(5) Disclosed herein are multimeric assemblies comprising a plurality of any one of the disclosed capsids within a membrane.

(6) Disclosed herein are modified non-enveloped viruses comprising a capsid wherein the capsid comprises a plurality of modified capsid proteins, wherein the plurality of modified capsid proteins comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the modified capsid protein, wherein the capsid forming protein is a capsid forming protein of a non-enveloped virus.

(7) Additional advantages of the disclosed method and compositions will be set forth in part in the description which follows, and in part will be understood from the description, or may be learned by practice of the disclosed method and compositions. The advantages of the disclosed method and compositions will be realized and attained by means of the elements and combinations particularly pointed out in the appended claims. It is to be understood that both the foregoing general description and the following detailed description are exemplary and explanatory only and are not restrictive of the invention as claimed.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The accompanying drawings, which are incorporated in and constitute a part of this specification, illustrate several embodiments of the disclosed method and compositions and together with the description, serve to explain the principles of the disclosed method and compositions.

(2) FIGS. 1A-C show enveloped, non-enveloped, and “quasi-enveloped” viruses. FIG. 1A shows enveloped viruses comprising a single cargo-containing capsid (orange) surrounded by membrane (green) derived from the host cell during budding. Embedded in the membrane are viral fusion proteins (red) that are required for entering target cells. FIG. 1B shows that non-enveloped viruses are released upon cell lysis and bind receptors directly with their capsid. Membrane-disrupting elements are released after endocytosis to enter the cytosol. FIG. 1C shows that quasi-enveloped viruses exit cells and acquire membrane by budding from producer cell. Often, multiple capsids are released inside single vesicles. Capsids are infectious with or without membranes.

(3) FIGS. 2A-C show the concept of enveloping non-enveloped virus capsids. FIG. 2A shows that Flock House Virus (FHV) and Nodamuravirus (NoV) capsid proteins Alpha serve as assembly domains. FIG. 2B shows that the membrane-binding PH domain from Phospholipase C (PLC) and ESCRT-recruiting p6^{sup}.Gag sequences were placed into surface loops of the proteins. FIG. 2C shows the resulting quasi-enveloped virus capsids.

(4) FIGS. 3A-C show the design of enveloped FHV (eFHV) and enveloped Nodamura virus NoV (eNoV). FIG. 3A shows the structure of FHV and NoV capsids (left). Schematic of modified proteins: p6Gag and PH domains were placed in two different positions in the Alpha protein to generate enveloped versions of Alpha (termed eAlpha). A Myc tag was added to detect the proteins with antibodies. FIG. 3B Modified FHV Alpha proteins are released. eAlpha proteins that contain all functional domains for membrane binding, assembly and ESCRT recruitment are released (lanes 2-4, and 6-8) while controls are not (lanes 1 and 5). When eAlpha proteins are present, Alpha proteins are released, indicating that both proteins form an assembly (lines 2, 3, 6, and 7). FIG. 3C Modified NoV Alpha proteins are released. eAlpha proteins that contain all functional domains for membrane binding, assembly and ESCRT recruitment are released (lanes 2-4, 6-8,) while controls are not (lanes 1 and 5). When eAlpha proteins are present, Alpha proteins are released, indicating that both proteins form an assembly (lines 2, 3, 6, and 7).

(5) FIGS. 4A-B show that eFHV and eNoV bud inside membrane envelopes and that the release is ESCRT-dependent. FIG. 4A depicts the results of a protease protection assay that shows that eFHV and eNoV are released inside a membrane: Lane 1 shows the released protein, lane 2 shows the released protein is protected after trypsin digestion, and lane 3 shows that addition of Triton to strip away the membrane leads to degradation of the protein by Trypsin. FIG. 4B shows that the vesicle release is ESCRT-dependent because overexpression of a dominant negative VPS4 enzyme completely blocks the release of eFHV and eNoV (lanes 4 and 8).

(6) FIGS. 5A-B show evidence that eFHV is released inside a vesicle. FIG. 5A shows a model of the eFHV capsid with the FHV capsid protein in orange and the PH domain in blue. ESCRT

binding sequences are unstructured and therefore not shown. FIG. 5B shows the icosahedral structure of the correct size (arrow) inside a vesicle membrane (arrowhead) shown by cryo-EM tomography.

(7) FIGS. 6A-B shows that eFHV can deliver cargo into target cells. FIG. 6A shows expression (lower panel) and release (upper panel) of eFHV, VSV-G, and GFP. Lane 1 shows the control, which was only transfected with GFP and VSV-G. Lane 2 shows expression of unmodified FHV Alpha together with eAlpha in a ratio 3:1 (note, Alpha is not detected on the western because it has no Myc-tag) together with GFP and VSV-G. Lane 3 shows expression of unmodified FHV Alpha together with eAlpha in a ratio 3:1 together with GFP and a mutant VSV-G that cannot mediate fusion with the target cell (P/D). Lane 4 shows expression of eAlpha together with GFP and VSV-G and lane 5 shows expression of eAlpha, GFP and mutant VSV-G. Note that in the presence of eAlpha, VSV-G and small amounts of GFP are also released into the supernatant. FIG. 6B shows delivery of GFP into target cells. eFHV can deliver GFP into target cells when co-expressed with VSV-G (lanes 2 and 4).

(8) FIG. 7 shows the design of enveloped adeno-associated virus (AAV), eAAV. FIG. 7A shows the structure of AAV capsids (top left) and a model of capsids packaged into a vesicle (bottom left). Schematic of designing eAAV: AAV capsids consist of VP1, VP2, and VP3 proteins, which originate from different start codons in the AAV genome (right, top panel). By mutating these start codons, expression of VP2 can be uncoupled from VP1 and VP3 (right, center panel). The N terminus of VP2, which is not required for AAV assembly or infectivity, can now be modified with p6.sup.Gag and the 13 N-terminal residues of Lyn kinase that contain an N-terminal myristoylation and palmitoylation sequence. A Myc tag was added to detect the proteins with antibodies.

DETAILED DESCRIPTION

(9) The disclosed method and compositions may be understood more readily by reference to the following detailed description of particular embodiments and the Example included therein and to the Figures and their previous and following description.

(10) It is to be understood that the disclosed method and compositions are not limited to specific synthetic methods, specific analytical techniques, or to particular reagents unless otherwise specified, and, as such, may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only and is not intended to be limiting.

(11) Disclosed are materials, compositions, and components that can be used for, can be used in conjunction with, can be used in preparation for, or are products of the disclosed method and compositions. These and other materials are disclosed herein, and it is understood that when combinations, subsets, interactions, groups, etc. of these materials are disclosed that while specific reference of each various individual and collective combinations and permutation of these compounds may not be explicitly disclosed, each is specifically contemplated and described herein. Thus, if a class of molecules A, B, and C are disclosed as well as a class of molecules D, E, and F and an example of a combination molecule, A-D is disclosed, then even if each is not individually recited, each is individually and collectively contemplated. Thus, in this example, each of the combinations A-E, A-F, B-D, B-E, B-F, C-D, C-E, and C-F are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. Likewise, any subset or combination of these is also specifically contemplated and disclosed. Thus, for example, the sub-group of A-E, B-F, and C-E are specifically contemplated and should be considered disclosed from disclosure of A, B, and C; D, E, and F; and the example combination A-D. This concept applies to all aspects of this application including, but not limited to, steps in methods of making and using the disclosed compositions. Thus, if there are a variety of additional steps that can be performed it is understood that each of these additional steps can be performed with any specific embodiment or combination of embodiments of the disclosed methods, and that each such combination is specifically contemplated and should be considered disclosed.

A. Definitions

(12) It is understood that the disclosed method and compositions are not limited to the particular methodology, protocols, and reagents described as these may vary. It is also to be understood that the terminology used herein is for the purpose of describing particular embodiments only, and is not intended to limit the scope of the present invention which will be limited only by the appended claims.

(13) It must be noted that as used herein and in the appended claims, the singular forms “a”, “an”, and “the” include plural reference unless the context clearly dictates otherwise. Thus, for example, reference to “a modified capsid protein” includes a plurality of such proteins, reference to “the modified capsid protein” is a reference to one or more modified capsid proteins and equivalents thereof known to those skilled in the art, and so forth.

(14) A “capsid forming protein” is a structural protein which is part of the complex forming a capsid. As used herein, a “capsid forming protein” refers to an amino acid sequence capable of forming a capsid. For example, a capsid forming protein can be an amino acid sequence responsible for the self-interaction between capsid proteins. In some aspects, a capsid forming protein can be an alpha protein or VP2. For example, a capsid forming protein can be an FHV Alpha protein or AAV VP2 protein.

(15) As used herein, “a capsid” is the protein shell of a virus. The capsid comprises one or more proteins referred to as capsid proteins or protomers. The capsid can enclose the viral genome or simply enclose nucleic acids, proteins, or small molecules for delivery. In some aspects, the capsid can have an icosahedral structure. In some aspects the capsid can be helical or prolate.

(16) As used herein, “a capsid protein” is a protein that when assembled with other capsid proteins forms a capsid. A capsid protein can also be known as a capsid subunit or substructure because it is a smaller element of the capsid. In some aspects, a capsid protein is referred to as a protomer.

(17) As used herein, “a modified capsid protein” is a capsid protein that is not naturally occurring or native to that particular capsid protein. A modified capsid protein can be a capsid protein with an amino acid substitution or an amino acid that has been altered from the amino acid sequence found in nature. A modified capsid protein can comprise sequences from other non-capsid proteins or from other non-heterologous capsid proteins. For example, a modified capsid protein can comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein.

(18) “Optional” or “optionally” means that the subsequently described event, circumstance, or material may or may not occur or be present, and that the description includes instances where the event, circumstance, or material occurs or is present and instances where it does not occur or is not present.

(19) Ranges may be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, also specifically contemplated and considered disclosed is the range from the one particular value and/or to the other particular value unless the context specifically indicates otherwise. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another, specifically contemplated embodiment that should be considered disclosed unless the context specifically indicates otherwise. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint unless the context specifically indicates otherwise. Finally, it should be understood that all of the individual values and sub-ranges of values contained within an explicitly disclosed range are also specifically contemplated and should be considered disclosed unless the context specifically indicates otherwise. The foregoing applies regardless of whether in particular cases some or all of these embodiments are explicitly disclosed.

(20) Unless defined otherwise, all technical and scientific terms used herein have the same

meanings as commonly understood by one of skill in the art to which the disclosed method and compositions belong. Although any methods and materials similar or equivalent to those described herein can be used in the practice or testing of the present method and compositions, the particularly useful methods, devices, and materials are as described. Publications cited herein and the material for which they are cited are hereby specifically incorporated by reference. Nothing herein is to be construed as an admission that the present invention is not entitled to antedate such disclosure by virtue of prior invention. No admission is made that any reference constitutes prior art. The discussion of references states what their authors assert, and applicants reserve the right to challenge the accuracy and pertinency of the cited documents. It will be clearly understood that, although a number of publications are referred to herein, such reference does not constitute an admission that any of these documents forms part of the common general knowledge in the art.

(21) Throughout the description and claims of this specification, the word “comprise” and variations of the word, such as “comprising” and “comprises,” means “including but not limited to,” and is not intended to exclude, for example, other additives, components, integers or steps. In particular, in methods stated as comprising one or more steps or operations it is specifically contemplated that each step comprises what is listed (unless that step includes a limiting term such as “consisting of”), meaning that each step is not intended to exclude, for example, other additives, components, integers or steps that are not listed in the step.

(22) The term “sequence of interest” or “gene of interest” can mean a nucleic acid sequence (e.g., a therapeutic gene), that is partly or entirely heterologous, i.e., foreign, to a cell into which it is introduced.

(23) The term “sequence of interest” or “gene of interest” can also mean a nucleic acid sequence, that is partly or entirely complementary to an endogenous gene of the cell into which it is introduced. For example, the sequence of interest can be micro RNA (miRNA), short hairpin RNA (shRNA), or short interfering RNA (siRNA).

(24) A “sequence of interest” or “gene of interest” can also include one or more transcriptional regulatory sequences and any other nucleic acid, such as introns, that may be necessary for optimal expression of a selected nucleic acid. A “protein of interest” means a peptide or polypeptide sequence (e.g., a therapeutic protein), that is expressed from a sequence of interest or gene of interest.

(25) The term “operatively linked to” refers to the functional relationship of a nucleic acid with another nucleic acid sequence. Promoters, enhancers, transcriptional and translational stop sites, and other signal sequences are examples of nucleic acid sequences operatively linked to other sequences. For example, operative linkage of DNA to a transcriptional control element refers to the physical and functional relationship between the DNA and promoter such that the transcription of such DNA is initiated from the promoter by an RNA polymerase that specifically recognizes, binds to and transcribes the DNA.

(26) Ranges can be expressed herein as from “about” one particular value, and/or to “about” another particular value. When such a range is expressed, another embodiment includes from the one particular value and/or to the other particular value. Similarly, when values are expressed as approximations, by use of the antecedent “about,” it will be understood that the particular value forms another embodiment. It will be further understood that the endpoints of each of the ranges are significant both in relation to the other endpoint, and independently of the other endpoint. It is also understood that there are a number of values disclosed herein, and that each value is also herein disclosed as “about” that particular value in addition to the value itself. For example, if the value “10” is disclosed, then “about 10” is also disclosed. It is also understood that when a value is disclosed that “less than or equal to” the value, “greater than or equal to the value” and possible ranges between values are also disclosed, as appropriately understood by the skilled artisan. For example, if the value “10” is disclosed then “less than or equal to 10” as well as “greater than or equal to 10” is also disclosed. It is also understood that throughout the application, data is provided

in a number of different formats, and that this data, represents endpoints and starting points, and ranges for any combination of the data points. For example, if a particular data point “10” and a particular data point “15” are disclosed, it is understood that greater than, greater than or equal to, less than, less than or equal to, and equal to 10 and 15 are considered disclosed as well as between 10 and 15.

B. Modified Capsid Protein

(27) Disclosed are modified capsid proteins comprising a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein.

(28) In some aspects, the capsid forming protein can be a capsid forming protein of an enveloped virus. Examples of enveloped virus include but are not limited to Herpesviridae, Poxviridae, Coronaviridae, Retroviridae, Orthomyxoviridae and Hepadnaviridae. In some aspects, the capsid forming protein can be a capsid forming protein of a non-enveloped virus. Examples of non-enveloped virus include but are not limited to Adenoviridae, Polyomaviridae, Papillomaviridae, Rudiviridae, Clavaviridae, Parvoviridae, Bimaviridae, Reoviridae, Totiviridae, Picomaviridae, Comoviridae, Bromoviridae, Hepeviridae, and Nodaviridae.

(29) In some aspects, the membrane binding element can be a membrane binding element of an enveloped virus. In some aspects, the ESCRT-recruiting element can be an ESCRT-recruiting element of an enveloped virus. In some aspects, the membrane binding element can be membrane binding element of a non-enveloped virus. In some aspects, the ESCRT-recruiting element can be an ESCRT-recruiting element of a non-enveloped virus.

(30) In some aspects, the membrane binding element can be a non-viral membrane binding protein. For example, the membrane binding element can be a cellular membrane binding protein. In some aspects, the ESCRT-recruiting element can be a non-viral membrane binding protein. In some aspects, the membrane binding element can comprise an amino acid sequence derived from a viral or non-viral membrane binding protein sequence.

(31) In some aspects, the membrane binding element can be the membrane binding PH element from phospholipase C. In some aspects, the ESCRT-recruiting element can be the p6^{sup}.Gag polypeptide from human immunodeficiency virus.

(32) In some aspects, the membrane binding element and/or the ESCRT-recruiting element does not occur in any naturally occurring protein (i.e. is non-naturally occurring). In some aspects, the membrane binding element and/or the ESCRT-recruiting element are synthetic. For example, the membrane binding element and/or the ESCRT-recruiting element can comprise an amino acid sequence derived from a viral or non-viral membrane binding protein sequence.

(33) In some aspects, the membrane binding element and the ESCRT-recruiting element can be a membrane binding element and an ESCRT-recruiting element of the same virus. In some aspects, the membrane binding element and the ESCRT-recruiting element can be a membrane binding element and an ESCRT-recruiting element of different viruses.

(34) In some aspects, the membrane binding element and ESCRT-recruiting element can be located within any region of the modified capsid protein that allows the capsid protein to retain its function and do not inhibit assembly of a capsid. For example, in some aspects the membrane binding element and ESCRT-recruiting element can be located within at least one exposed surface loop of the modified capsid proteins. In some aspects, the membrane binding element and ESCRT-recruiting element can be located within the same exposed surface loop of the modified capsid protein. In some aspects, the membrane binding element and ESCRT-recruiting element can be located within different exposed surface loops of the modified capsid protein. In some aspects, the membrane-binding element and ESCRT-recruiting element can be located at either the N or C terminus of the modified capsid protein. In some aspects, the membrane-binding element can be located at either the N or C terminus of the modified capsid protein and the ESCRT-recruiting element can be located at the opposite terminus of the modified capsid protein from the membrane-

binding element.

(35) In some aspects, the disclosed modified capsid proteins can further comprise a desired cargo, sequence of interest, gene of interest, packaging moiety, or a targeting moiety.

(36) 1. Membrane Binding Element

(37) The membrane binding element, also referred to as an “M domain,” can be any suitable polypeptide that is capable of binding to a lipid bilayer via any suitable mechanism, including but not limited to non-covalently interacting with the lipid bilayer membrane. In some aspects, such interactions can include but are not limited to interacting via specific binding pockets with the polar head groups of lipid molecules in the lipid bilayer, interacting electrostatically with charged polar head groups, interacting non-covalently with the hydrophobic interior of the lipid bilayer, or by harboring a chemical modification (non-limiting examples can be fatty acid or acylation modifications such as myristoylation) that interacts non-covalently with the lipid bilayer. A given membrane binding element can employ one or more mechanisms of interaction with a lipid bilayer. As described herein, the multimeric assembly described herein can comprise one or more membrane binding elements. In some aspects, each modified capsid protein in a multimeric assembly comprises one or more membrane binding elements. In other aspects, some fraction (30%, 40%, 50%, 60%, 70%, 80%, 90%, or more) of the plurality of modified capsid proteins comprise one or more membrane binding elements. In some aspects, each capsid protein in the plurality of modified capsid proteins comprises one or more membrane binding elements. In some aspects, one or more membrane binding elements is required per multimeric assembly as described herein in order to drive association of the multimeric assembly with the lipid bilayer via any suitable mechanism.

(38) The membrane binding elements present in a resulting modified capsid protein, capsid, multimeric assembly, or modified non-enveloped virus can all be the same, all different, or some the same and some different.

(39) In various embodiments, the one or more membrane binding elements can comprise or consist of a polypeptide having an acylation motif, including but not limited to N-terminal myristoylation motifs (including but not limited to MGXXXT/S (SEQ ID NO: 1) motif and non-limiting example sequences below), palmitoylation motifs (including but not limited to non-limiting example sequences below), farnesylation motifs, and geranylgeranylation motifs (Resh M (1999) Fatty acylation of proteins: new insights into membrane targeting of myristoylated and palmitoylated proteins. *Biochim. Biophys. Acta* 1451: 1-16; Resh M (2013) Covalent lipid modifications of proteins. *Curr. Biol.* 23:R431-5); a polar headgroup-binding domain (including but not limited to non-limiting example sequences 100-106 in the attached appendices and the domains defined in: Stahelin R V (2009) Lipid binding domains: more than simple lipid effectors. *J. Lipid Res.* 50:S299-304); or transmembrane protein domains (the latter preferably when the multimeric assembly is enveloped by a lipid bilayer). In various further embodiments, the M domain may comprise envelope proteins of enveloped viruses, membrane protein transporters, membrane protein channels, B-cell receptors, T-cell receptors, transmembrane antigens of human pathogens, growth factors receptors, G-protein coupled receptors (GPCRs), or complement regulatory proteins including but not limited to CD55 and CD59.

(40) In further embodiments, the membrane binding element may comprise or consist of one or more of the following peptides:

(41) TABLE-US-00001 (SEQ ID NO: 2) (M)GARAS; (Myr1; 6 N-terminal residues of HIV gag) (SEQ ID NO: 3) (M)GAQFS; (Myr2; 6 N-terminal residues of MARCKS) (SEQ ID NO: 4) (M)GSSKS; (Myr3; 6 N-terminal residues of Src) (SEQ ID NO: 5) (M)GKQNS; (Myr4; 6 N-terminal residues of Neurocalcin) (SEQ ID NO: 6) (M)GCIKSKRKDNLN; (Palm1; 13 N-terminal residues of Lyn kinase) (SEQ ID NO: 7) (M)GCTLSAEERAAL; (Palm2; 13 N-terminal residues of Gao) (SEQ ID NO: 8) (M)LC CMRRTKQ VEK; (Palm3; 13 N-terminal residues

of GAP43)(SEQ ID NO: 9)(M)DCLCIVTTKKYR;(Palm4; 13 N-terminal residues of PSD-95)(SEQ ID NO: 10)KKKKK SKTKC VIM;(CaaX1; 13 C-terminal residues from K-Ras4B)(SEQ ID NO: 11)DMKKHRCKCCSIM;(CaaX2; 13 C-terminal residues from paralemmin)(SEQ ID NO: 12)AQRQKKRRLCLLL;(CaaX3; 13 C-terminal residues of RhoF)(SEQ ID NO: 13)AQEFHQFLCNPL;(CaaX4; 13 C-terminal residues of type II inositol 1,4,5-trisphosphate 5-phosphatase isoform X7)(SEQ ID NO: 14)

HGLQDDPDLQALLKGSQLLKVKSSSWRRERFYKLQEDCKTIWQESRKVMR
SPESQLFSIEDIQEVRMGRTEGLEKFARDIPEDRCFSIVFKDQRNTLDL

IAPSPADAQHWVQGLRKIIHHSGSMDQRQK;(PH; Residues 11-40 of rat PLC5)(SEQ ID NO: 15)

PHRFKVHNYMSPTFCDHCGSLLWGLVKQGLKCEDCGMNVHHKCREKVAN LCG;(CI; residues 246-297 of human PCK5 isoform X2)(SEQ ID NO: 16)

GAVKLSVSYRNGTLFIMVMHIKDLVTEDGADPNPYVKTYLLPDTHKTSKR
KTKISRKTRNPTFNEMLVYSGYSKETLRQRELQLSVLSAESLRENFFLGG

ITLPLKDFNLSKETVKWYQLTAATYL;(C2; residues 1384-1509 of mouse PI3K) and/or (SEQ ID NO: 17)

AVAQQLRAESDFEQLPDDVAISANIADIEEKRGFTSHFVFVIEVKTKGGS KYL

IYRRYRQFH ALQ SKLEERF GPD SK S SAL AC TLPTLP AK V Y

VGVKQEI AEMRIPALN A Y MK SLLSLP VWVLMDED VRIFF YQ SP YD
SEQ VPQ ALRR(PX; residues 2-149 of human p40phox)

(42) Further exemplary membrane binding elements can comprise or consist of one or more of the peptides that follow (Resh M (1999) Biochim. Biophys. Acta 1451: 1-16; Resh M (2013) Curr. Biol. 23:R431-5; Stahelin RV (2009) J. Lipid Res. 50:S299-304).

(43) In some aspects, the membrane binding element can comprise the membrane binding myristoylated and palmitoylated element of Lyn kinase.

(44) i. N-Terminal Membrane Binding Elements

(45) The following peptides can be at the N terminus of the polypeptide in which they appear in order to function as a membrane binding element.

(46) Any amino acid sequence conforming to the consensus motif (M)GXXX(S/T) (SEQ ID NO: 1), where the membrane binding element is in the initiator methionine at the N terminus of the polypeptide sequence.

(47) TABLE-US-00002 (SEQ ID NO: 2)(M)GARAS (SEQ ID NO: 18)

(M)GCIKSKGKDSLS (SEQ ID NO: 19)(M)GCINSKRKD (SEQ ID NO: 20)

(M)GS SK SKPKDP SQRRR (SEQ ID NO: 21)(M)GCIKSKEDKGPAMKY (SEQ

ID NO: 52)(M)GCVQCKDKEATKLTE (SEQ ID NO: 53)(M)GCIKSKRKDNLNDDE

(SEQ ID NO: 54)(M)GCVCSSNPEDDWMEN (SEQ ID NO: 55)

(M)GCMKSKFLQVGGNTG (SEQ ID NO: 56)(M)GCVFCKKLEPVATAK (SEQ ID

NO: 57)(M)GCVHCKEKISGKGQG (SEQ ID NO: 58)(M)GLLSSKRQVSEKGGK

(SEQ ID NO: 59)(M)GQQPGKVLGDQRRPS (SEQ ID NO: 60)

(M)GQQVGRVGEAPGLQQ (SEQ ID NO: 61)(M)GNAAA AAKKGSEQESV (SEQ ID

NO: 62)(M)GNAATAKKGSEVESV (SEQ ID NO: 63)(M)GAQLSLVVQASPSIA

(SEQ ID NO: 64)(M)GHALCVCSRGTVIID (SEQ ID NO: 65)(M)GQLCCFPF

SRDEGKI (SEQ ID NO: 66)(M)GNEASYPLEMCSHFD (SEQ ID NO: 67)

(M)GNSGSKQHTKHNSKK (SEQ ID NO: 68)(M)GCTLSAEDKAAVERS (SEQ ID

NO: 69)(M)GCTLSAEERAALERS (SEQ ID NO: 70)(M)GAGASAEKHSRELE

(SEQ ID NO: 71)(M)GCRQSSEEKEAARRS (SEQ ID NO: 72)

(M)GLSFTKLFSRLFAKK (SEQ ID NO: 73)(M)GNIFGNLLKSLIGKK (SEQ ID

NO: 74)(M)GLTVSALFSRIFGKK (SEQ ID NO: 75)(M)GKVL SKIFGNKEMRI (SEQ

ID NO: 76)(M)GNSKSGALSKEILEE (SEQ ID NO: 77)(M)GKQNSKLRPEVMQDL

(SEQ ID NO: 78) (M)GKRASKLKPEEVEEL (SEQ ID NO: 79)
 (M)GKQNSKLRPEVLQDL (SEQ ID NO: 80) (M)GSRASTLLRDEELEE (SEQ ID NO: 81) (M)GSKLSKKGKGYNVND (SEQ ID NO: 82) (M)GKQNSKLRPEMLQDL (SEQ ID NO: 83) (M)GNVMEGKSVEELSST (SEQ ID NO: 84) (M)GQQF SWEEAEENGAV (SEQ ID NO: 85) (M)GNTKSGALSKEILEE (SEQ ID NO: 86) (M)GKQNSKLRPEVLQDL (SEQ ID NO: 87) (M)GAQF SKTAAKGEATA (SEQ ID NO: 88) (M)GSQSSKAPRGDVTAE (SEQ ID NO: 89) (M)GNRHAK ASSPQGFDV (SEQ ID NO: 90) (M)GQDQTKQQIEKGLQL (SEQ ID NO: 91) (M)GQALSIKSCDFHAAE (SEQ ID NO: 92) (M)GNRAFKAHNGHYLSA (SEQ ID NO: 93) (M)GARASVLSGGELDRW (SEQ ID NO: 94) (M)GQTVTTPLSLTLTDHW (SEQ ID NO: 95) (M)GQAVTTPL SLTLTDHW (SEQ ID NO: 96) (M)GNSPSYNPPAGISPS (SEQ ID NO: 97) (M)GQTLTTPLSLTLTHF (SEQ ID NO: 98) (M)GQTITTPL SLTLTDHW (SEQ ID NO: 99) (M)GQTVTTPLSLTLEHW (SEQ ID NO: 100) (M)GQELSQHERRYVEQLK (SEQ ID NO: 101) (M)GVSGSKGQKLFVSVL (SEQ ID NO: 102) (M)GGKWSKSSVVGWPTV (SEQ ID NO: 103) (M)GQHPAKSMDVRRIEG (SEQ ID NO: 104) (M)GAQVSRQNVGTHSTQ (SEQ ID NO: 105) (M)GLAFSGARPCCCRHN (SEQ ID NO: 106) (M)GNRGSSTSSRPPLSS (SEQ ID NO: 107) (M)GSYFVPPANYFFKDI (SEQ ID NO: 108) (M)GAQLSTLSRVVLSVP (SEQ ID NO: 109) (M)GNLKSVMGQEPGPPCG (SEQ ID NO: 110) (M)GSKRSVPSRHRSLTT (SEQ ID NO: 111) (M)GNGESQLSSVPAQKL (SEQ ID NO: 112) (M)GAHLVRRYLGDASVE (SEQ ID NO: 113) (M)GGKL SKKKKGYNVND (SEQ ID NO: 114) (M)GSCCSCPDKDTVPDN (SEQ ID NO: 115) (M)GS SEVSIIPGLQKEE (SEQ ID NO: 116) (M)LCCMRRTKQ VEKNDE (SEQ ID NO: 117) (M)GCLGNSKTEDQRNE (SEQ ID NO: 118) (M)TLESIMACCLSEEAKEA (SEQ ID NO: 119) (M)SGVVRTLSRCLLPAAEAG (SEQ ID NO: 120) (M)ADFLPSRSVCFPGCVLTN (SEQ ID NO: 121) (M)ARSLRWRC CPWCLTEDEK AA (SEQ ID NO: 122) (M)LCCMRRTKQVEKNDDQKIEQDGI (SEQ ID NO: 123) (M)QCCGLVHRRRV (SEQ ID NO: 124) (M)DCLCIVTTKKYRYQDEDTP (SEQ ID NO: 125) (M)CKGLAGLPASCLRSAKDMK (SEQ ID NO: 126) (M)GCIKSKEDKGPAMKY (SEQ ID NO: 127) (M)GCVQCKDKEATKLTE (SEQ ID NO: 128) (M)GCIKSKRKDNLNDE (SEQ ID NO: 129) (M)GCVCSSNPEDDWMEN (SEQ ID NO: 130) (M)GCMKSKFLQVGGNTG (SEQ ID NO: 131) (M)GCVFCKKLEPVATAK (SEQ ID NO: 132) (M)GCVHCKEKISGKGQG (SEQ ID NO: 133) (M)GCTLSAEDKAAVERS (SEQ ID NO: 134) (M)GCTLSAEERAALERS (SEQ ID NO: 135) (M)GCRQ SSEEKEAARRS (SEQ ID NO: 136) (M)GQLCCFPFSRDEGK (SEQ ID NO: 137) (M)GNLKSVMGQEPGPPCGLGLGLGLGLCGK

ii. C-Terminal Membrane Binding Elements

(48) The following peptides can be at the C terminus of the polypeptide which they appear in order to function as a membrane binding element.

(49) TABLE-US-00003 (SEQ ID NO: 138) SGPGCMSCKCVLS (SEQ ID NO: 139) GTQGCMGLPCVVM (SEQ ID NO: 140) TPGCVKIKKCVIM (SEQ ID NO: 141) DMKKHRCKCCSIM (SEQ ID NO: 142) SKDGKKKKK SKTKCVIM (SEQ ID NO: 143) KKKKKKSKTKC (SEQ ID NO: 144) SKTKCVIM

iii. Polar Headgroup Binding Domains that Function as Membrane Binding Elements

(50) The following peptides are non-limiting examples of polar headgroup-binding domains that can function as membrane binding elements. These domains can appear anywhere in the polypeptides of the invention consistent with proper folding and multimerization of the multimeric assembly.

(51) TABLE-US-00004 (SEQ ID NO: 145)

HGLQDDPDQLQALLKVKSSSWRRERFYKLQEDCKTIWQESRKVMR
 SPESQLFSIEDIQEVRMIGHRTEGLEKFARDIPEDRCFSIVFKDQRNTLDL
 IAPSPADAQHWVQGLRKIIHHSMDQRQK (SEQ ID NO: 146)
 (M)DSGRDFLTLLHGLQDDPDQLALLKGSQLLKVKSSSWRRERF YKLQED
 CKTIWQESRKVMRSPESQLF SIEDIQEVRMIGHRTEGLEKF ARDIPEDR CF
 SIVFKDQRNTLDLIAPSPADAQHWVQGLRKIIHHSMDQRQK (SEQ ID NO: 147)
 (M)DSGRDFLTLLHGLQDDPDQLALLKGSQLLKVKSSSWRRERF YKLQED
 CKTIWQESRKVMRSPESQLF SIEDIQEVRMIGHRTEGLEKF ARDIPEDR CF
 SIVFKDQRNTLDLIAPSPADVQHWVQGLRKIIDRSGSMDQRQK (SEQ ID NO: 148)
 (M)DSGRDFLTLLHGLQDDEDLQ ALLKGSQLLKVKSSSWRRERFYKLQED
 DCKTIWQESRKVMRTPESQLF SIEDIQEVRMIGHRTEGLEKF ARDVPED RCF
 SIVFKDQRNTLDLIAPSPADAQHWVLGLHKIIHHSMDQRQK (SEQ ID NO: 149)
 HGLQDDEDLQ ALLKGSQLLKVKSSSWRRERF YKLQEDCKTIWQESRK
 VMRTPESQLFSIEDIQEVRMIGHRTEGLEKFARDVPEDRCFSIVFKDQRNT
 LDLIAPSPADAQHWVLGLHKIIHHSMDQRQK (SEQ ID NO: 150)
 (M)SGGKYVDSEGHLYTVPIREQGNIYKPNK AMAEEMNEKQ VYD AH
 TKEIDLNVNRDPKHLNDDVVKIDFEDVIAEPEGTHSFDGIWKASFTTFTVT
 KYWFYRLLSALFGIPMALIWGIYFAILSFLHIWAVVPCIKSFLIEIQCIS
 RVYSIYVHTFCDPLFEAIGKIF SNIRINTQKEI (SEQ ID NO: 151) (M)SGGKYVD
 SEGHLYTVPIREQGNIYKPNKAMADELSEKQ VYDAHT
 KEIDLNVNRDPKHLNDDVVKIDFEDVIAEPEGTHSFDGIWKASFTTFTVTK
 YWFYVpLLSALFGIPMALIWGIYFAILSFLHIWAVVPCIKSFLIEIQCIS
 RVYSIYVHTVCDPLFEAVGKIF SNVRINLQKEI

(52) Based on the disclosure herein, it is well within the level of those of skill in the art to identify membrane binding elements suitable for use in producing the modified capsid proteins, capsids, multimeric assemblies, and modified non-enveloped viruses disclosed herein. In one embodiment, a suitable membrane binding element can be identified as follows: As described throughout, a membrane binding element for use in the present disclosure can be any suitable polypeptide element that is capable of binding to a lipid bilayer via any suitable mechanism, including but not limited to non-covalently interacting with the lipid bilayer membrane. As will be known to those of skill in the art, a membrane binding element can be demonstrated to perform the function of membrane binding using a variety of standard assays. Many in vitro assays exist for assaying whether or not a polypeptide interacts with lipid membranes and for evaluating the characteristics of the interaction, such as the nature of the interaction (e.g., electrostatic or hydrophobic), the strength of the interaction, and whether the interaction deforms or remodels the membrane. Such assays include but are not limited to vesicle sedimentation assays, vesicle co-flotation assays, isothermal titration calorimetry, measuring changes in intrinsic or extrinsic protein or lipid fluorescence, fluorescence anisotropy, and membrane morphology analysis by electron microscopy or fluorescence microscopy (Zhao H, Lappalainen P (2012) A simple guide to biochemical approaches for analyzing lipid-protein interactions. *Mol. Biol. Cell* 23:2823-30). In addition, membrane binding element-dependent localization of proteins to membranes in cells can also be used as an assay for the interaction of a membrane binding element with membranes, and can yield information about the specificity of a given M domain for particular membranes, membrane subdomains, or lipids (Zacharias D A, Violin J D, Newton A C, Tsien R Y (2002) Partitioning of lipid-modified GFPs into membrane microdomains in live cells. *Science* 296:913-916; Lemmon M A (2008) Membrane recognition by phospholipid-binding domains. *Nat. Rev. Mol. Cell. Biol.* 9:99-111). Whether in vitro or in cells, either an isolated membrane binding element or a membrane binding element linked via genetic fusion or another method to a carrier protein that facilitates observation (for example, green fluorescent protein) can be used to evaluate the ability of the membrane binding element to interact with lipid membranes.

(53) 2. ESCRT-Recruiting Element

(54) The ESCRT-recruiting element, also referred to as an “L domain,” can be used in any suitable polypeptide that is capable of effecting membrane scission by recruiting the ESCRT machinery to the site of budding by binding to one or more ESCRT or ESCRT-associated proteins directly or indirectly via any suitable mechanism, including but not limited to non-covalently or covalently. Preferably, the ESCRT-recruiting element interacts with proteins known to recruit the ESCRT machinery to sites of budding in vivo, such as Tsg101, ALIX, or the Ned4 family of ubiquitin E3 ligases (McDonald B, Martin-Serrano J (2009) No strings attached: the ESCRT machinery in viral budding and cytokinesis. *J. Cell Sci.* 122:2167-77; Votteler J, Sundquist W I (2013) Virus budding and the ESCRT pathway. *Cell Host & Microbe* 14:232-41). Most preferably, the ESCRT-recruiting element interacts with the human, murine, or other mammalian forms of these proteins. Each protein subunit in a multimeric assembly contains one or more L domains. The ESCRT-recruiting elements present in a resulting multimeric assembly may all be the same, all different, or some the same and some different.

(55) In various embodiments, the one or more ESCRT-recruiting element can comprise or consist of a linear amino acid sequence motif selected from the group consisting of P(T/S)AP (SEQ ID NO: 152), Φ YX.sub.0/2(P/ Φ)X.sub.0/3(I/T), PPXY (SEQ ID NO: 153), and overlapping combinations thereof (Bieniasz P D (2006) Late budding domains and host proteins in enveloped virus release. *Virology* 344:55-63; Votteler J, Sundquist W I (2013) Virus budding and the ESCRT pathway. *Cell Host & Microbe* 14:232-41), where Φ denotes a hydrophobic residue, X can be any amino acid, and numbered subscripts indicate amino acid spacers of varying lengths. Such overlapping combinations include, but are not limited to P(T/S)APPXY (SEQ ID NO: 155), P(T/S)APYP(X)_nL (SEQ ID NO: 156), PPXYP(T/S)AP (SEQ ID NO: 157), PPXYYP(X).sub.nL (SEQ ID NO: 158), YP(X).sub.nLPPXY (SEQ ID NO: 159), and YP(X).sub.nLPPXY (SEQ ID NO: 160).

(56) Further exemplary ESCRT-recruiting element can comprise or consist of one or more of the peptides that follow:

(57) TABLE-US-00005 (SEQ ID NO: 161) PTAPPEE; (SEQ ID NO: 162) YPLTSL; (SEQ ID NO: 163) PTAPPEY; (SEQ ID NO: 164) YPDL; (SEQ ID NO: 165) FPIV; (SEQ ID NO: 166) PTAPPEY; (SEQ ID NO: 167) PTAP; (SEQ ID NO: 168) PPEY; (SEQ ID NO: 169) YPLTSL; and/or (SEQ ID NO: 165) FPIV.

(58) As will be understood by those of skill in the art, the ESCRT-recruiting element can include additional sequences, beyond those directly responsible for recruiting the ESCRT machinery, as appropriate for an intended use, so long as the ESCRT-recruitment motifs are not buried in the peptide core in such a way as to render them inaccessible for binding their interaction partners. In various further embodiments, the ESCRT-recruiting element can comprise or consist of the following peptides that include one or more ESCRT-recruitment motifs plus additional residues (ESCRT-recruitment motifs noted by underlined text):

(59) TABLE-US-00006 (SEQ ID NO: 172) LQSRPE PTAPPEE SFRSGVETTTTPPKQKQEPIDKEL YPLTSLRSLFGNDPSSQ; (HIV Gag p6 domain) (SEQ ID NO: 173) RRVLPTAPPEYMEAIYPVR; (residues 2-21 of Ebola VP40) (SEQ ID NO: 174) PIQKKSQHNSVQETPQTQNLYPDLSEIKKEYNVKEKDQVEDLNLDLSEWE; (EIAV Gag p9 domain) (SEQ ID NO: 175) NPRQ SIKAFPIVINSDGGEK; (residues 12-31 of SV5 M) (SEQ ID NO: 176) PTAPPEYGGS; (SEQ ID NO: 177) PTAPGGS; (SEQ ID NO: 178) PPEYGGS; (SEQ ID NO: 179) YPLTSLGGS; (SEQ ID NO: 180) YPDLGGS; (SEQ ID NO: 181) FPIVGGS; (SEQ ID NO: 182) LQSRPEAAAPEESFRSGVETTTTPPKQKQEPIDKELYPLTSLRSLFGNDPSSQ, (HIV Gag p6 domain mutant (SEQ ID NO: 183) p(APTAP)); and/or (SEQ ID NO: 184) LOSRPE PTAPPEE SFRSGVETTTTPPOKOEPIDKELAAALTSLSLFGNDPSSQ, (HIV

Gag p6 domain mutant p6(Δ YP))

(60) Further exemplary ESCRT-recruiting elements can comprise or consist of one or more of the following polypeptides:

(61) TABLE-US-00007 (SEQ ID NO: 186)

QSRPEPTAPPEESFRSGVETTTTPPQKQEPIDKELYPLTSLRSLFGNDP SSQ (SEQ ID NO: 187) DPQIPPPPYVEPTAPQV (SEQ ID NO: 188) LLTEDPPPYRD (SEQ ID NO: 154)

TASAPPPPYVG (SEQ ID NO: 171) TPQTQNLYPDLSSEIK (SEQ ID NO: 170)

(M)RRVILPTAPPEYMEAI (SEQ ID NO: 51) NTYMQYLNPPPYADHS (SEQ ID

NO: 50) LGIAPPPYEEDTSMEYAPSAP (SEQ ID NO: 49) DDLWLPPPEYVPLKEL

(SEQ ID NO: 48) AAPTAPPTGAADSIPPPYSP (SEQ ID NO: 47) TAPSSPPPYEE

(SEQ ID NO: 46) QSIKAFPIVINS DG (SEQ ID NO: 185)

SREKPYKEVTEDLLHLNSL (SEQ ID NO: 22) AAGAYDPARKLLEQYAKK (SEQ ID

NO: 23) PNCFNSSINNIHEMEIQLKDALEKNQQWLVDQDQREVYVKGLLAKIFE

LEKKTETAHSLPQQTKKPESEGYLQEEKQKC (SEQ ID NO: 24)

RKSPTPSAPVPLTEPAAQ (SEQ ID NO: 25)

(M)SLYPSLEDLKVDKVIQAQTAFSANPANPAILSEASAPIPHDGNLY

PRLYPELSQYMGLSLN

(62) Based on the disclosure herein, it is well within the level of those of skill in the art to identify ESCRT-recruiting element suitable for use in producing the modified capsid proteins, multimeric assemblies, and modified non-enveloped viruses disclosed herein. As described herein, aESCRT-recruiting element for use in the present invention can be any suitable polypeptide domain that is capable of effecting membrane scission and release of an enveloped multimeric assembly from a cell by recruiting the ESCRT machinery to the site of budding by binding to one or more ESCRT proteins directly or indirectly via any suitable mechanism, including but not limited to non-covalently or covalently. As will be known to those with skill in the art, the ability of an ESCRT-recruiting element to recruit the ESCRT machinery and effect membrane scission and release of an enveloped multimeric assembly can be assessed using budding assays. In the budding assay, a candidate ESCRT-recruiting element is genetically fused to a viral structural protein that has been rendered defective in budding by mutation or deletion of its late domain, and the ability of the candidate ESCRT-recruiting element to restore budding of virus-like particles is evaluated by analyzing the culture supernatant for the presence of the viral structural protein using standard techniques such as SDS-PAGE and Western blotting (Parent L J, Bennett R P, Craven R C, Nelle T D, Krishna N K, Bowzard J B, Wilson C B, Puffer B A, Montelaro R C, Wills J W (1995) Positionally independent and exchangeable late budding functions of the Rous Sarcoma Virus and Human Immunodeficiency Virus Gag proteins. *J. Virol.* 69:5455-5460). Any viral structural protein that is known to be defective in budding can be used in the budding assay, including but not limited to budding-defective versions of HIV-1 Gag, RSV Gag, MuMoLV Gag, SV5 M, Ebola VP40 and other structural proteins from different families of enveloped viruses including retroviruses, filoviruses, rhabdoviruses, arenaviruses, and paramyxoviruses. In addition, as described below, the multimeric assemblies of the invention can be used to test the ability of an L domain to effect membrane scission and release of an enveloped multimeric assembly in a similar manner. The ESCRT-recruiting element of a modified capsid protein can be replaced with a candidate ESCRT-recruiting element, and the ability of the resulting construct to be released from cells can be determined by analyzing the culture supernatant for the presence of the protein subunits of the multimeric assembly using standard techniques such as SDS-PAGE and Western blotting. Finally, as will be known to those with skill in the art, the ability of an ESCRT-recruiting element to bind to one or more ESCRT proteins directly or indirectly can be assessed using a variety of biochemical, biophysical, and cell biological techniques including but not limited to co-immunoprecipitation, pull-down assays, isothermal titration calorimetry, biosensor binding assays, NMR spectroscopy, and X-ray crystallography.

(63) 3. Cargo

(64) In some aspects, the disclosed modified capsid proteins can further comprise a desired cargo. The desired cargo can be, but is not limited to, a nucleic acid (e.g. a sequence of interest or a gene of interest), protein (e.g. a peptide of interest or a protein of interest), a ribonucleoprotein complex, a small molecule, or any combination thereof.

(65) In some aspects, a desired cargo can be anything of interest that can be enclosed within a capsid or found on the surface of a capsid. In some aspects, a desired cargo can be linked to or interact with a packaging moiety disclosed herein and thus recruited to the multimeric assembly, including but not limited to therapeutics, diagnostics, antigens, adjuvants, imaging agents, dyes, radioisotopes, etc. In some aspects, if the desired cargo is a protein or polypeptide, the cargo can be expressed as a genetic fusion with the membrane binding element, or the ESCRT-recruiting element in order to directly incorporate the desired cargo into a modified capsid protein or multimeric assembly without the use of a distinct packaging moiety. In various embodiments, the desired cargo can be selected from the group consisting of, but not limited to, proteins, nucleic acids, lipids, small organic compounds or combinations thereof.

(66) In some aspects, the cargo can be a detectable label.

(67) In some aspects, the desired cargo comprises a polynucleotide with a nucleic acid sequence which is a recognition sequence known to bind to the corresponding packaging moieties described herein. In some aspects, the polynucleotide with a nucleic acid sequence which is a recognition sequence known to bind to a packaging moiety can be

(68) TABLE-US-00008 (SEQ ID NO: 26)

GGUCUGGGCGCACUUCGGUGACGGUACAGGCC (Ig70 RNA sequence) (SEQ ID NO: 27) AAUCCAUUGCACUCCGGAUUU (ula RNA sequence) (SEQ ID NO: 28) GGCGACUGGUGAGUACGCCAAAAUUUUGACUAGCGGAGGCUAG (HIV_NC RNA sequence) (SEQ ID NO: 29) GGCUCGUGUAGCUCAUUAGCUCCGAGCC (lmb RNA sequence)

(69) In some aspects, the capsid protein is translated from a nucleic acid sequence that itself is replicated by the viral enzymes together with the nucleic acid sequence encoding the cargo. In this case, both the polynucleotide sequences encoding the capsid as well as the cargo contain recognition sequences for the viral replication enzymes. In some aspects, the nucleic acid sequence to bind a viral replication enzyme can be:

(70) TABLE-US-00009 (SEQ ID No 30)

GTAAACAATTCCAAGTTCCAAAATGGTTAATAACAACAGACCAAGACGTC
AACGAGCTCAACGCGTTGTCTGCACACAACCCAAACAGCGCCTGTTCCA
CAGCAAAACGTGCCACGTAATGGTAGACGCCGACGTAATCGCACGAGGCG
TAATCGCCGACGTGTGCGCGGAATGAACATGGCGGCGCTAACCAGATTAA
GTCAACCTGGTTTGGCGTTTCTCAAATGTGCATTTGCACCACCTGACTTT
CATTCAGGTACGCTTCCATGAACGTGGGTATTTACCCAACGTTCGAAGTTG
ATGCAGTTTGCCGGAAGCATAACTGTTTGGAATGCCCTGTAAAGCTGAG
TACTGTGCAATTCCCGGTTGCAACAGATCCAGCCACCAGTTCGCTAGTTC
ATACTCTTGTTGGTTTAGATGGTGTCTAGCGGTGGGGCCTGACAACTTC
TCTGAGTCATTCATGAAGGATTTGGCTTTTAGAAGCATCCGGACGCCAAC
CTAACCGGGCAAGTATCCGAACAATCGGACATTTGGCCACAATAAGCCCA
ATTTGGTTGAAGATTAAAGTAGTGAGCCCCCTTAGCGCGAAACCGGAATT
TATATTCCAAACCAGTTTAAGTCAACAGACTAAGGT (Flock House Virus defective
interfering (DI)-RNA sequence).

(71) In some aspects, the desired cargo comprises a nucleic acid comprising one or more of the disclosed recognition sequences. Examples can be found in International Application Publication No WO 2016/138525, herein incorporated by reference in its entirety.

(72) 4. Packaging Moiety

(73) In a further embodiment, the capsids, multimeric assemblies, or modified non-enveloped viruses of any embodiment or combination of embodiments herein can further comprise a packaging moiety. As used herein, a “packaging moiety” can be any moiety capable of interacting with a desired “cargo”, with the effect of recruiting the cargo to the capsid, multimeric assembly, or modified non-enveloped virus. The interaction between the packaging moiety and the cargo can be any type of interaction, covalent or non-covalent, that results in effective interaction with and recruitment to the capsid, multimeric assembly, or modified non-enveloped virus. As will be apparent to those of skill in the art, the ability to widely modify surface amino acid residues without disruption of the protein structure permits many types of modifications to endow the resulting self-assembled multimers with a variety of functions. In one non-limiting example, at least one of the modified capsid proteins can be modified, such as by introduction of various cysteine residues or non-canonical amino acids at defined positions to facilitate linkage to one or more cargo of interest. In another non-limiting example, the modified capsid protein can be modified to comprise as a genetic fusion a polypeptide domain or sequence known to interact with a desired cargo covalently or non-covalently. In one embodiment, a non-canonical amino acid can be incorporated recombinantly using amber codon suppression (see L. Wang, A. Brock, B. Herberich, P. G. Schultz, Science 292, 498 (2001)). In another embodiment, the packaging moiety comprises the polypeptide sequence:

(74) TABLE-US-00010 (SEQ ID NO: 186)

QSRPEPTAPPEESFRSGVETTTTPPQKQEPIDKELYPLTSLRSLFGNDP SSQ,

wherein the packaging moiety polypeptide is expressed as a genetic fusion with the M domain or the L domain. This sequence is the p6 domain of HIV Gag, which is known to interact with the HIV protein Vpr via a non-covalent protein-protein interaction (Cavrois M, et al. (2002) Nat. Biotech. 20: 1151-4). For example, by including SEQ ID NO: 186 in a multimeric assembly of the invention, any polypeptide sequence or other molecule that is fused, tethered, or otherwise connected to the Vpr sequence can be packaged into the multimeric assembly.

(75) In some aspects, the packaging moiety can be bound or fused to the membrane binding element, the ESCRT-recruiting element, or the capsid-forming protein.

(76) Additional packaging moieties can comprise or consist of one or more of the following peptides expressed as a genetic fusion with the membrane binding element, the ESCRT-recruiting element, or the capsid-forming protein, each of which binds to corresponding recognition sequences present in a nucleic acid cargo of interest, resulting in recruitment of the nucleic acid cargo of interest to the multimeric assembly.

(77) TABLE-US-00011 (SEQ ID NO: 31) (a) DRRRRGSRPSGAERRRRRAAAA(Ig70) (SEQ ID NO: 32) (b) AVPETRPNHTIYINNLNEKIKKDELKKSLHAIFSRFGQILDILVS RSLKMRGQAFVIFKEVSSATNALRSMQGFPFYDKPMRIQYAKTDSIIA KMK (ula)

(SEQ ID NO: 33) (c)

(M)QKGNFRNQRKTVKCFNCGKEGHIAKNCRAPRKKGCWKCGKEG

HQMKDCTERQAN, (HIV NC) and/or (SEQ ID NO: 34) (d) RPRGTRGKGRRIIR

(mnb) (SEQ ID NO: 35) (e) MVNNNRPRRQRAQRVVVTTTQTAPVPQQNV (FHV Alpha-31) (SEQ ID NO: 36) (f)

MTLKVILGEHQITRTELLVGIATVSGCGAVVYCISKFWGYGAIAP

YPQSGGNRVTRALQRAVIDKTKTPIETRFYPLDSLRTVTPKRVADNGHA

VSGAVRDAARRLIDESITAVGGSKFEVNPNSSTGLRNHFHFAVGDLA

QDFRNDTPADDAFIVGVDVDYYVTEPDVLEHMRPVVLHTFNPKKVSGF

DADSPFTIKNNLVEYKVSGGAAWVHPVWDWCEAGEFIASRVRTSWKEWF

LQLPLRMIGLEKVG YHKIHHCRPWTDCPDRLVYTIPQYVIWRFNWIDT

ELHVRKLKRIEYQDET KPGWNRLEYVTDKNELLVSIGREGEHAQITIEK

EKLDMLSGLSATQSVNARLIGMGHKDPQYTS MIVQYYTGKKVVSPIST

VYKPTMPRVHWPVTSDADVPEVSARQYTLPIVSDCMMMPMIKRWETMSE

SIERRVTFVNSDRIAKIAETFKVLMNGPFKDLDPKLSIIEETIERL
 NKPSQQQLRAVFEMIGVKPRQLIESFNKNEPGMKSSRIISGFPDILFI
 LKVSRYTLAYSIVLHAEHNEHWYYPGRNPTEIADGVCEVSDCDAEVI
 ETDFSNLDGRVSSWMQRNIAQKAMVQAFRPEYRDEIISFMDTIINCPAK
 AKRFGFRYEPGVGVKSGSPTTTPHNTQYNGCVEFTALTFEHPDAEPEDL
 FRLIGPKCGDDGLSRAIIQKSINRAAKCFGLELKVERYNPEIGLCFLSR
 VFVDPLATTTTIQDPLRTLRLHLTLTRDPTIPLADAACDRVEGYLCTDA
 LTPLISDYCKMVLRLYGPTASTEQVRNQRRSRNKEKPYWLTCDGSWPQH
 PQDAHLMKQVLIKRTAIDEDQVDALIGRFAAMKDVWEKITHDSEESAAA
 CTFDEDGVAPNSVDESPLMLNDAKQTRANPGTSRPHSNGGGSSHGNEP
 RRTEQRAQGPRQPARLPKQGKTNGKSDGNITAGETQRRGGIPRGKGRGG
 KTNRTRTPPKAGAQPQPSNNRKLEKLASRSEQKLISEEDL (FHV Protein A)

5. Additional Moieties

(78) In some aspects, the disclosed capsids, multimeric assemblies, and modified non-enveloped viruses further comprise one or more transmembrane protein or membrane-anchored protein embedded in the lipid bilayer. This embodiment can be used to add additional functionality of any desired type to the capsids, multimeric assemblies, and modified non-enveloped viruses. In this embodiment, the transmembrane protein or membrane-anchored protein can be one not present as part of the capsid or modified capsid proteins, in that they are added to the assembly or virus during or after envelopment of the multimeric assembly and modified non-enveloped virus by the lipid bilayer and do not necessarily interact with the protein subunits either covalently or non-covalently. Any suitable transmembrane protein or membrane-anchored protein can be added that provides any desired additional functionality to the assembly, in terms of cell targeting, the display of transmembrane or membrane-anchored antigen for vaccines, or other desired use. In one non-limiting example, the transmembrane protein or membrane-anchored protein embedded in the lipid bilayer comprises a viral envelope protein that enables the enveloped multimeric assembly, capsid or modified non-enveloped virus to enter cells via receptor-mediated endocytosis and/or mediates fusion of the lipid bilayer of the enveloped multimeric assembly, capsid or modified non-enveloped virus with cellular membranes. In the study of enveloped viruses, the practice of incorporating a foreign viral envelope protein in the membrane of an enveloped virus is referred to as “pseudotyping.” By co-expressing the foreign viral envelope protein with the viral or virus-like particle proteins, the foreign viral envelope protein becomes embedded in the membrane bilayer of the cells, and is therefore incorporated into the membrane envelope of the budding virions or virus-like particles. As the inventors have shown below, viral envelope proteins (in one embodiment, the G protein of Vesicular Stomatitis Virus) can be incorporated in the membrane envelopes of the enveloped multimeric assemblies, capsids or modified non-enveloped viruses of the disclosure in a similar manner. In various non-limiting embodiments, additional classes of membrane proteins can be incorporated into the membrane envelopes of the multimeric assemblies, capsid or modified non-enveloped viruses of the disclosure. In various non-limiting embodiments, the transmembrane or membrane-anchored protein is selected from the group consisting of the envelope proteins of enveloped viruses, membrane protein transporters, membrane protein channels, B cell receptors, T cell receptors, transmembrane antigens of human pathogens, growth factors receptors, G-protein coupled receptors (GPCRs), complement regulatory proteins including but not limited to CD55 and CD59, or processed versions thereof.

(79) In specific embodiments, the one or more transmembrane protein or membrane-anchored protein embedded in the lipid bilayer comprise one or more of the following polypeptides, or a processed version thereof. As will be understood by those of skill in the art, the polypeptide sequences provided are full-length protein precursors, which are cleaved or otherwise processed (i.e., “processed”) to generate the final envelope protein embedded in the lipid bilayer.

(80) TABLE-US-00012 VSV-G (SEQ ID NO: 37)

MKCLLYLFLVFNHKNVPSNHYCPSSSDLNWHN
DLIGTALQVKMPKSHKAIQADGWMCHASKWVTTCDFRWYGPKYITHSIRS
FTPSVEQCKESIEQTKQGTWLNPGFPPQSCGYAWTDAEAVIVQVTPHHVL
VDEYTG EWVDSQFINGKCSNYICPTVHNSTTWHS DYKVKGLCDSNLISMD
ITFFSEDGELSSLGKEGTGFRSNYFAYETGGKACKMQYCKHWGVRLPSGV
WFEMADKDLFAAARFPECPEGSSISAPSQTSVDVSLIQDVERILDYSLCQ
ETWSKIRAGLPISPVDLSYLAPKNPGTGPAFTIINGTLKYFETRYIRVDI
AAPILSRMVGMISGTTTERELWDDWAPYEDVEIGPNGVLRTSSGYKFPLY
MIGHGMLDSDLHLSSKAQVFEHPHIQDAASQLPDDES LFFGDTGLSKNPI
ELVEGW FSSWKSSIASFFFIIGLIIGLFLVLRVGIHL CIKLKHTIKKRQI YTDIEMNRLGK;
Ecotropic envelope protein from Moloney Murine Leukemia Virus or "Fco" (SEQ
ID NO: 38) MARSTLSKPLKNKNVNPRGPLIPIALLMLRGVSTASPGSSPFIQVYNITWE
VINGDRET VWATSGNHPLWTWWPDLTPLDCLMLAHHGPSYWGLEYQSPFSS
PPGPPCCSGGSSPGCSRDC EEPLTSLTPRCNTAWNRLKLDQTTHKSNEGF
YVCPGPHRPRESKSCGGPDSFYCAYWGGETTGRAYWKPSSSWDFITVNNN
LTSDQAVQVCKDNKWCNPLVIRFTDAGRRVTSWTTGHYNWGLRLYVSGQD
PGLTFGIRLRYQNLGPRVPIGPNPVLADQQPLSKPKPKVKS P SVTKPPSGT
PLSPTQLPPAGTENRLLNLVDGAYQALNLTSPDKTQECWLCTVAGPFYYE
GVAVLGTYSNHTSAPANCSVASQHKLTLSEVTGQGLCIGAVPKTHQALCN
TTQTSSRGSYYL VAPTGTMWACSTGLTPCISTTILNLTTDYCVLVELWPR
VTYHSPSYVYGLFERSNRHKREPVS LTLALLGGLTMGGIAAGIGTGTTA
LMATQQFQQLQAAVQDDLREVEKSISNLEKSLTSLSEVVLQNRRLDLLF
LKEGGLCAALKEECFYADHTGLVRDSMAKL RERLNQRQKLFESTQGWTE
GLFNRSPWF T TLISTIMGPLIVLLMILLFGPCILNRLVQFVKDRISVVQA
LVL TQQYHQLKPIEYEP; Amphotropic Murine Leukemia Virus Envelope 4070A
(SEQ ID NO: 39)

MARSTLSKPPQDKINPWKPLIVMGVLLGVGMAESPHQVFNV TWRVTNLMT
GRTANATSL LGTVQDAFPKLYFDL CDLVGEEWDPSDQEPYVGYGCKYPAG
RQRTRTFDFYVCPGHTVKSGCGGPGE GYCGKWGCETT GQAYWKPTSSWDL
ISLKRGNTPWDTGCSKVACGPCYDLSKVSNSFQGATRGGRCNPLVLEFTD
AGKKANWDGPKSWGLRLYRTGTD PITMFSLTRQVLNVGPRVPIGPNPVL P
DQRLPSSPIEIVPAPQPPSPLNTSYPPSTTSTPSTSPSPVPQPPPGTG
DRLLALVKGAYQALNLTNPDKTQECWLCLVSGPPY YEGVAVVGTYTNHST
APANCTATSQHKLTLSEVTGQGLCMGAVPKTHQALCNTTQSAGSGSYLA
APAGTMWACSTGLTPCLSTTVLNLTTDYCVLVELWPRVIYHSPD MYGQL
EQRTKYKREPVS LTLALLGGLTMGGIAAGIGTGTTALIKTQQFEQLHAA
IQTDLNEVEKSITNLEKSLTSLSEVVLQNRRLDLLFLKEGGLCAALKEE
CCFYADHTGLVRDSMAKL RERLNQRQKLFETGQGWFEGLFNRSPWF T TLI
STIMGPLIVLLLILLFGPCILNRLVQFVKDRISVVQALVL TQQYHQLKPI EYEP; Sindbis
virus E3-E2-6K-E1 envelope polyprotein (SEQ ID NO: 40) SAAPLVTAMCLLGNV
SFPCDRPPTCYTREP SRALDILEENVNHEAYDTLLNAILRCGS
SGRSKRSVIDDFTLTSPYLG TCSYCHHTVPCFSPVKIEQVWDEADDNTIR
IQTSAQFGYDQSGAASANKYRYMSLKQDHTVKEGTMD DIKISTSGPCRRL
SYKGYFLLAKCPPGDSVTVSIVSSNSATSCTLARKIKPKFVGREKYDLPP
VHGKKIPCTVYDRLKETTAGYITMHRPRPHAYTSYLEESSGKVYAKPPSG
KN_nYECKCGDYKTGTVSTRTEITGCTAIKQCVAYKSDQTKWVFNSPD LIR
HDDHTAQGKLHLPFKLIPSTCMVPVAHAPNVIHGFKHISLQLD TDHLLTLL
TTRRLGANPEPTTEWIVGKWRNFWD RDGLEYIWGNHEPVRVYAQESAPGD
PHGWPHEIVQHYYHRHPWTILAVASAWAMMIGVT VAVLCACKARRECLTP
YALAPNAVIPTSLALLCCVRSANAETFTETMSYLWSNSQPFFWVQLCIPL

AAFIVLMRCCSSCLPFLVAYFLAHATWPNVVPQIPYKALVERA
GYAPLNLEITVMSSEVLPSTNQEYITCKFTTVVPSPKIKCCGSLECPAA
HADYTCKVFGGVYPFMWGGAQCFCDSSENSQMSEAYVELSADCASDHAQAI
KVHTAAMKVGLRIVYGNTTSFLDVYVNGVTPGTSKDLKVIAGPISASFTP
FDHKVVIHRGLVYNYDFPEYGAMKPGAFGDIQATSLTSKDLIASTDIRLL
KPSAKNVHVPYTQASSGFEMWKNSGRPLQETAPFGCKIAVNPLRAVDCS
YGNIPISIDIPNAAFIRTSAPLVSTVKCEVSECTYSADFGGMATLQYVS
DREGQCPVHSHSSTATLQESTVHVLEKGAVTVHFSTASPQANFIVSLCGK
KTTCNAECKPPADHIVSTPHKNDQEFQAAISKTSWSWLFAFGGASSLLI
IGLMIFACSMMLTSTRR; Ebola GP (Zaire Mayinga strain) (SEQ ID NO: 41)
MGVTGILQLPRDRFKRTSFFLWVILFQRTFSIPLGVIHNSTLQVSDVDK
LVCRDKLSSTNQLRSVGLNLEGNGVATDVPSATKRWGFRRSGVPPKVNYE
AGEWAENCYNLEIKKPDGSECLPAAPDGIRGFPRCRYVHKVSGTGPCAGD
FAFHKEGAFFLYDRLASTVIYRGTTFAEGVVAFLILPQAKKDFSSHPLR
EPVNATEDPSSGYYSTTIRYQATGFGTNETEYLFEDNLTIVQLESRFTP
QFLLQLNETIYTSGKRSNTTGKLIWKVNPEIDTTIGEWAFWETKKNLTRK
IRSEELSFTVVSNGAKNISGQSPARTSSDPGTNTTTEDHKIMASENSSAM
VQVHSQGREAAVSHLTTLATISTSPQSLTTKPGPDNSTHNTPVYKLDISE
ATQVEQHHRRTDNDSTASDTPSATTAAAGPPKAENTNTSKSTDFLDPATTT
SPQNHSETAGNNNTHHQDTGEESASSGKLGLITOTIAGVAGLITGGRRTO
REAIVNAQPKCNPNLHYWTTQDEGAAIGLAWIPYFGPAAEGIYIEGLMHN
QDGLICGLRQLANETTQALQLFLRATTELRTFSILNRKAIDFLLQRWGGT
CHILGPDCCIEPHDWTKNITDKIDQIIHDFVDKTLDPDQGDNDNWWWTGWRQ
WIPAGIGVTGVIIAVIALFCICKFVF; Human Immunodeficiency Virus envelope
glycoprotein precursor gp160 (SEQ ID NO: 42)
MRVKEKYQHLWRWGWKKGIMLLGILMICSATENLWVWYYGVPVWKEATTT
LFCASDAKAYDTEViiNVCATHACVPTDPNPQEVILNVNVTENFDMWKNDM
VEQMHEDIISLWDQSLKPCVKLTPLCVNLKCTDLKNDTOTOSSNGRMIME
KGEIKNCSFNISTSIRNKVQKEYAFFYKLDIRPIDNTTYRLISCNTSVIT
QACPKVSFEPIPIHYCAPAGFAILKCNDKTFNGTGPCTNVSTVQCTHGIR
PVVSTQLLLNGSLAEEEGVIRSANFTDNAKTIIVQLNTSVEINCTRPNNN
TRKSIRIQRGPGRAFVTIGKIGNMRQAHCNISRAKWMSTLKQIASKLREQ
FGNNKTVIFKQSSGGDPEIVTHSFNCGGEFFYCNSTQLFNSTWFNSTWST
EGSNNTGSDTITLPCRIKQFINMWQEVGKAMYAPPISGQIRCSSNITGL
LLTRDGGKNTNESEVFRPGGGDMRDNRSELYKYKVVKIETLGVAPTKAK
RRVVQREKRAVGIGALFLGLGAAGSTMGAASMTLTVQARQLLSGIVQQQ
NNLLRAIEAQQHLLQLTVWGIKQLQARILAVERYLKDQQLLGIWGCSSGKL
ICTTAVPWASWSNKSLEQFWNNMTWMEWDREINNYTSLIHSLIDESQNQQ
EKNEQELLELDKWASLWTATMITTALWIKIFIMIVGGLVGLRIVFAVLSI
VNRVRQGYSPLSFQTHLPNRGGPDRPEGIEEEGGERDRDRSVRLVNGSLA
LIWDDLRLSLCLFSYHRLRDLILLIVTRIVELLGRRGWEALKYWWNLLQYWS
QELKNSAVSLLNATAIAVAEGTDRVIEVVQGAYRAIRHIPRRIRQGLERI L; Respiratory
Syncytial Virus F protein precursor (SEQ ID NO: 43)
MELLILKANAITTILTAVTFCFASGQNITEEFYQSTCSAVSKGYLSALRT
GWYTSVITIELSNIKENKCNGTDAKVKLIKQELDKYKNAVTELQLLMQST
PPTONRARRELPRFMNYTLNNAKKTNTVLSKKRKRRFLGFLLGVSIAIAS
GVAVSKVLHLEGEVNKIKSALLSTNKAVVSLSNGVSVLTSKVLDLKNYID
KQLLPIVNKQSCSISNIETVIEFQQKNNRLLLEITREFSVNAGVTTPVSTY
MLTOSELLSLINDMPITNDQKKLMSNNVQIVRQQSYSIMSIIKEEVLAYV
VQLPLYGVIDTPCWKLHTSPLCTTNTKEGSNICLTRTRDRGWYCDNAGSVS

FFPQAETCKVKQSNRVFCDTTVINSLTLPSEINLCNVDFNPKYDCKIMTS
KTDVSSSVITSLGAIVSCYGKTKCTASNKNRGIKTFNSNGCDYVSNKGM
TVSVGNTLYYVVKQEGKSLYVKGEPIINFYDPLVFPSPDEFDASISQVNEK
INQSLAFIRKSDELLHNVNAGKSTTNIMITTHIIVILLSLIAVGLLL
YCKARSTPVTLSKDQLSGINNIAFSN; SARS Coronavirus spike protein (SEQ ID
NO: 44) MFIFLLFLTLTSGSDLDRCTTFDDVQAPNYTQHTSSMRGVYYPDEIFRSD
TLYLTQDLFLPFYSNVTGFHTINHTFGNPVIPFKDGIYFAATEKSNVVRG
WVFGSTMNNKSQSVIIINNSTNVVIRACNFELCDNPFFAVSKPMGTQTH
MIFDNAFNCTFEYISDAFSLDVSEKSGNFKHLREFVFKNKDGLYVYKGY
QPIDVVRDLPSGFNTLKPFIKFLPLGINITNFRAILTAFSPAQDIWGTSAA
AYFVGYLKPTTFMLKYDENGTTITDAVDCSQNPLAELKCSVKSFIDKGIY
QTSNFRVVPSPGDVVRFPNITNLCPFGEVFNATKFPSVYAWERKKISNCVA
DYSVLYNSTFFSTFKCYGVSATKLNDLCFSNVYADSFVVKGDDVRQIAPG
QTGVADIYNYKLPDDFMGCVLAWNTRNIDATSTGNYNYKYRYLRHKGKLRP
FERDISNVPFSPDGKPCPTPALNCYWPLNDYGFYTTTGIGYQPYRVVLS
FELLNAPATVCGPKLSTDLIKNCVNFNFNGLTGTGVLTPSSKRFQPFQ
FGRDVSDFTDSVRDPKTSEILDISPCSFSGGVSVITPGTNASSEVAVLYQD
VNCTDVSTAIHADQLTPAWRIYSTGNNVFQTQAGCLIGAEHVDTSYEC
PIGAGICASYHTVSLRSTSQKSIVAYTMSLGADSSIAYSNNTIAIPTNF
SISITTEVMPVSMAKTSVDCNMYICGDSSTECANLLLQYGSFCTQLNRALS
GIAAEQDRNTREVFAQVKQMYKTPTLKYFGGFNFSQILPDPLKPTKRSFI
EDLLFNKVTLADAGFMKQYGECLGDINARDLICAQKFNGLTVLPPLTDD
MIAAYTAALVSGTATAGWTFGAGAALQIPFAMQMAYRFNGIGVTQNVLYE
NQKQIANQFNKAISQIQESLTTTSTALGKLQDVVNQNAQALNTLVKQLSS
NFGAISSVLNDILSRLDKVEAEVQIDRLITGRLQSLQTYVTQQLIRAAEI
RASANLAATKMSECVLGQSKRVDFCGKGYHLMSPQAAPHGVVFLHVTYV
PSQERNFTTAPAICHEGKAYFPREGVFVFNGTSWFITQRNFFSPQIITD
NTFVSGNCDVVIGIINNTVYDPLQPELDSFKEELDKYFKNHTSPDVDLGD
ISGINASVVNIQKEIDRLNEVAKNLNESLIDLQELGKYEQYIKWPWYVWL
GFIAGLIAIVMVTILLCCMTSCCSCCLKGACSCGSCCKFDEDDSEPVKGV KLHYT;

Influenza hemagglutinin (SEQ ID NO: 45)

MKTIIALSYIFCLALGQDLPGNDNSTATLCLGHHAVPNGTLVKTITDDQI EVTNATELVQ
SSSTGKICNNPHRILDGIDCTLIDALLGDPHCDVFQNETWDLFVERSKAF SNCYPYDVDP
YASLRSLVASSGTLEFITEGFTWTGVTQNGGSNACKRGP GSGFFSRLNWL
TKSGSTYPVLNVTMPNNDNFDKLYIWGIHHPSTNQEQTSLYVQASGRVTV
STRRSQQTIIPNIGSRPWVRGLSSRISYWTIVKPGDVLVINSNGNLIAP
RGYFKMRTGKSSIMRSDAPIDTCISECITP
NGSIPNDKPFQNVNKnYGACPKYVKQNTLKLATGMRNVPEKQTRGLFGAI
AGFIENGWEGMIDGWYGRHQNSEGTGQAADLKSTQAAIDQINGKLN RVI
EKTNEKFHQIEKEFSEVEGRIQDLEKYVEDTKIDLWSYNAELLVALENQH
TIDLT DSEMKNLFEKTRRQLRENAEEMGNCGFKIYHKCDNACIESIRNGT
YDHDVYRDEALNNRFQIKGVELKSGYKDWILWISFAISCFLLCVVLLGFI
MWACQRGNIRCNICI.

(81) In some aspects, any known targeting moiety can be used. A targeting moiety can be any nucleic acid or amino acid sequence that targets a particular cell surface protein or intracellular target.

(82) As used herein, the term “targeting moiety” refers to any moiety that specifically binds a selected target. The targeting moiety can be, for example, a polysaccharide, a peptide, peptide ligand, an aptamer, an antibody or fragment thereof, a single chain variable fragment (scFv) of an antibody, or a Fab' fragment, or a nanobody. Targeting moieties can also include other forms of an

antibody as disclosed in Rissiek et al.; “Nanobodies as modulators of inflammation: potential applications for acute brain injury,” *Front. Cell. Neurosci.*, 21 Oct. 2014; and Cuesta et al.; “Multivalent antibodies: when design surpasses evolution,” *Trends in Biotechnology*; Vol. 28, Issue 7, pp. 355-362, July 2010. The cited references are incorporated herein by reference in their entirety.

(83) As used herein, a “targeting moiety” can be specific to a recognition molecule on the surface of a cell or a population of cells, such as, for example B cells or T cells. In an aspect of the disclosed compositions and methods, a targeting moiety can include, but is not limited to: a monoclonal antibody, a polyclonal antibody, full-length antibody, a chimeric antibody, Fab’, Fab, F(ab).sub.2, F(ab’).sub.2, a single domain antibody, Fv, a single chain Fv (scFv), a minibody, a diabody, a triabody, hybrid fragments, a phage display antibody, a ribosome display antibody, an oligonucleotide, a modified oligonucleotide, a peptide, a peptide ligand, a hormone, a growth factor, a cytokine, a saccharide or polysaccharide, and an aptamer.

(84) As used herein, “aptamers” refer to molecules that interact with a target molecule, preferably in a specific way. Typically, aptamers are small nucleic acids ranging from 15-50 bases in length that fold into defined secondary and tertiary structures, such as stem-loops or G-quartets. Aptamers can bind small molecules and large molecules. Aptamers can bind very tightly with Kd's from the target molecule of less than 10.sup.-12 M. Aptamers can bind the target molecule with a very high degree of specificity. Aptamers are known to the art and representative examples of how to make and use aptamers to bind a variety of different target molecules can be found in the following non-limiting list of U.S. Pat. Nos. 5,476,766, 5,503,978, 5,631,146, 5,731,424, 5,780,228, 5,792,613, 5,795,721, 5,846,713, 5,858,660, 5,861,254, 5,864,026, 5,869,641, 5,958,691, 6,001,988, 6,011,020, 6,013,443, 6,020,130, 6,028,186, 6,030,776, and 6,051,698. In an aspect, the aptamer can be synthetic, nonimmunogenic antibody mimics. In an aspect, the aptamer can be a DNA aptamer. In an aspect, the DNA aptamer can be anti-PD-1 aptamer.

(85) As used herein, the term “contacting” refers to bringing a disclosed composition, compound, conjugate or protein together with an intended target (such as, e.g., a cell or population of cells, a receptor, an antigen, or other biological entity) in such a manner that the disclosed composition, compound, conjugate or fusion protein can affect the activity of the intended target (e.g., receptor, transcription factor, cell, population of cells, etc.), either directly (i.e., by interacting with the target itself), or indirectly (i.e., by interacting with another molecule, co-factor, factor, or protein on which the activity of the target is dependent). In an aspect, a disclosed composition or fusion protein can be contacted with a cell or population of cells, such as, for example, one or more lymphocytes (e.g., T cells and/or B cells).

C. Capsid

(86) Disclosed are capsids comprising a plurality of the modified capsid proteins disclosed herein. Thus, for example, disclosed are capsids comprising a plurality of modified capsid proteins, wherein the modified capsid proteins comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein.

(87) In some aspects, the modified capsid proteins are derived from a non-envelope virus. In other words, the modified capsid proteins can comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein and wherein the capsid forming protein can be a capsid forming protein of a non-enveloped virus. If the capsid forming protein is from a non-enveloped virus then the capsid can be referred to as a non-enveloped capsid.

(88) In some aspects, the plurality of modified capsid proteins interact with each other forming the capsid.

(89) In some aspects, the membrane binding element and ESCRT-recruiting element can be located

within a region of the plurality of modified capsid proteins that does not disrupt the ability of the capsid to form.

(90) In some aspects, the disclosed capsids can further comprise a packaging moiety as described herein. The packaging moiety can be present on one or more of the modified capsid proteins. In some aspects, the packaging moiety is bound to a desired cargo. For example, the packaging moiety can be bound to a desired cargo with a covalent or non-covalent bond. In some aspects, the packaging moiety can be a modified amino acid within one or more of the modified capsid proteins. In some aspects, the packaging moiety can be a polypeptide.

(91) In some aspects, the disclosed capsids further comprise a desired cargo as described herein.

(92) In some aspects, the disclosed capsids can further comprise a membrane surrounding the capsid. The membrane can be a lipid bilayer. Thus, the presence of a membrane surrounding a capsid comprised of modified capsid proteins, wherein the modified capsid proteins comprise capsid forming proteins of a non-enveloped virus can convert a non-enveloped capsid to an enveloped capsid.

D. Multimeric Assembly

(93) Disclosed are multimeric assemblies comprising a one or more of the capsids disclosed herein within a membrane. Disclosed are multimeric assemblies comprising a two or more of the capsids disclosed herein within a membrane. Thus, for example, disclosed are multimeric assemblies comprising a plurality of capsids, wherein the capsids comprise a plurality of modified capsid proteins, wherein the modified capsid proteins comprise a capsid forming protein, a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein.

(94) In some aspects, the membrane binding element and ESCRT-recruiting element can be located within a region of the plurality of modified capsid proteins that does not disrupt the ability of the multimeric assembly to form.

(95) As disclosed herein, the membrane can be a lipid bilayer. As used herein, the membrane can also be referred to as an envelope.

(96) In some aspects, the multimeric assembly further comprises a packaging moiety. In some aspects, the packaging moiety can be bound to a desired cargo. In some aspects, the packaging moiety is bound to a desired cargo with a covalent or non-covalent bond. In some aspects, the packaging moiety can be a modified amino acid within one or more of the modified capsid proteins. In some aspects, the packaging moiety can be a polypeptide. In some aspects, the desired cargo can be a nucleic acid, protein, or small molecule. The packaging moiety and desired cargo can be any of those disclosed herein.

E. Modified Non-Enveloped Virus

(97) Disclosed herein are modified non-enveloped viruses comprising one or more of the capsids disclosed herein. Thus, for example, disclosed are modified non-enveloped viruses comprising a capsid wherein the capsid comprises a plurality of modified capsid proteins, wherein the plurality of modified capsid proteins comprise a membrane binding element and an ESCRT-recruiting element, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the modified capsid protein, wherein the capsid forming protein is a capsid forming protein of a non-enveloped virus.

(98) In some aspects, the membrane binding element and ESCRT-recruiting element can be located within any region of the modified capsid protein that they retain their function and do not inhibit assembly of a capsid. For example, in some aspects the membrane binding element and ESCRT-recruiting element can be located within at least one exposed surface loop of the modified capsid proteins. In some aspects, the membrane binding element and ESCRT-recruiting element can be located within the same exposed surface loop of the modified capsid protein. In some aspects, the membrane binding element and ESCRT-recruiting element can be located within different exposed surface loops of the modified capsid protein. In some aspects, the membrane-binding element and

ESCRT-recruiting element can be located at either the N or C terminus of the modified capsid protein. In some aspects, the membrane-binding element can be located at either the N or C terminus of the modified capsid protein and the ESCRT-recruiting element can be located at the opposite terminus of the modified capsid protein from the membrane-binding element.

(99) In some aspects, the modified non-enveloped virus can be, but is not limited to, Adenoviridae, Polyomaviridae, Papillomaviridae, Rudiviridae, Clavaviridae, Parvoviridae, Bimaviridae, Reoviridae, Totiviridae, Picomaviridae, Comovirinae, Bromoviridae, Hepeviridae, and Nodaviridae. Thus, the capsid forming protein can be a capsid forming protein of, but is not limited to, Adenoviridae, Polyomaviridae, Papillomaviridae, Rudiviridae, Clavaviridae, Parvoviridae, Bimaviridae, Reoviridae, Totiviridae, Picomaviridae, Comovirinae, Bromoviridae, Hepeviridae, or Nodaviridae.

(100) In some aspects, the modified non-enveloped virus can be a Flock House Virus or Nodamura Virus. In some aspects, one or both of the membrane binding element and ESCRT-recruiting element are located at position 206 of the alpha protein of Flock House Virus. In some aspects, one or both of the membrane binding element and ESCRT-recruiting element are located at position 264 of the alpha protein of Flock House Virus. In some aspects, one or both of the membrane binding element and ESCRT-recruiting element can be located at position 199 of the alpha protein of Nodamura Virus. In some aspects, one or both of the membrane binding element and ESCRT-recruiting element can be located at position 259 of the alpha protein of NoV. In some aspects, membrane binding and ESCRT recruiting elements can be located at the N terminus of the VP2 protein of AAV. In some aspects, membrane binding and ESCRT recruiting elements can be located at the truncated N terminus of the VP2 protein of AAV.

(101) In some aspects, the membrane binding element can be any of the membrane binding elements described herein. For example, the membrane binding element can be the membrane binding PH element from phospholipase C.

(102) In some aspects, the ESCRT-recruiting element can be any of the ESCRT-recruiting elements described herein. For example, the ESCRT-recruiting element can be the p6.sup.Gag polypeptide from human immunodeficiency virus.

(103) In some aspects, the disclosed modified non-enveloped viruses can further comprise a membrane surrounding the capsid. In some aspects, the membrane can be a lipid bilayer. In some aspects, the membrane can also be referred to as an envelope. Thus, the presence of the membrane surrounding the disclosed modified non-enveloped viruses can convert the modified non-enveloped viruses to enveloped viruses and can be referred to as quasi-enveloped viruses.

(104) In some aspects, the membrane can comprise any of the targeting moieties described herein.

(105) In some aspects, the disclosed modified non-enveloped viruses can be replication defective.

(106) In some aspects, the disclosed modified non-enveloped viruses can comprise a desired cargo. The desired cargo can be any of those disclosed herein. In some aspects, the desired cargo can be the viral genome. In some aspects, the viral genome can be modified.

F. Nucleic Acid Sequences

(107) Disclosed are nucleic acid sequences that encode any of the disclosed modified capsid proteins.

(108) Disclosed are constructs comprising the nucleic acid sequence that encodes any of the modified capsid proteins disclosed herein.

(109) Disclosed are constructs comprising a modified viral genome wherein the modified viral genome comprises a nucleic acid sequence capable of encoding any of the disclosed modified capsid proteins. In some aspects, the modified viral genome can be a modified Flock House Virus or Nodamuravirus genome. In some aspects, the modified viral genome can be a modified AAV genome.

(110) In some aspects, a modified viral genome is a viral genome that renders the resulting virus replication defective.

G. Methods of Producing

(111) Disclosed are methods of producing a vector comprising transfecting a cell with a packaging plasmid comprising a sequence of interest; and any of the constructs disclosed herein. In some aspects, the sequence of interest can be a therapeutic agent or a detectable agent.

H. Methods of Treating

(112) Disclosed are methods of treating a subject with any of the disclosed capsids, any of the disclosed multimeric assemblies, any of the disclosed modified non-enveloped virus, or any of the disclosed vectors produced by the methods disclosed herein.

(113) In some aspects, the sequence of interest encodes a protein that is non-functional or absent in the subject. In some aspects, the sequence of interest can be a therapeutic agent.

I. Kits

(114) The materials described above as well as other materials can be packaged together in any suitable combination as a kit useful for performing, or aiding in the performance of, the disclosed method. It is useful if the kit components in a given kit are designed and adapted for use together in the disclosed method. For example disclosed are kits for producing the disclosed modified capsid proteins, capsids, multimeric assemblies or modified non-enveloped viruses, the kit comprising a membrane binding element and an ESCRT-recruiting element. The kits also can contain lipids, desired cargo, or targeting moieties.

EXAMPLES

(115) One goal of the current disclosure is to generate a virus-based vector system in which capsids from non-enveloped viruses are modified so that they efficiently bud from cells inside membrane enclosed vesicles, thereby converting non-enveloped viruses into quasi-enveloped viruses. Three modular activities are necessary: membrane binding, self-assembly, and the ability to recruit ESCRT machinery to catalyze the membrane fission reactions required for vesicle release.

(116) Quasi-enveloped viral capsids were generated by modifying the capsid-forming Alpha proteins from two non-enveloped nodaviruses: Flock House (FHV) and Nodamuraviruses (NOV). The membrane-binding PH domain from phospholipase C (PH-PLC) and the ESCRT-recruiting p6 sequence from HIV-1 Gag were inserted within exposed surface loops of the capsid-forming Alpha protein (FIG. 2).

(117) Quasi-enveloped viral capsids were generated by modifying the VP2 protein in the capsid of Adeno-associated virus. The membrane-binding 13 N-terminal amino acids of Lyn kinase, which include a myristoylation and a palmitoylation motif, and the ESCRT-recruiting p6 sequence from HIV-1 Gag, were inserted at the N terminus of the VP2 protein (FIG. 7).

(118) 1. Plasmid Design and Construction

(119) The polypeptide sequences of the ESCRT-recruiting HIV-1 p6Gag domain and the membrane-binding PH domain were inserted into exposed surface loops of the Alpha proteins (at amino acid positions 264 and 206 of FHV Alpha, and at equivalent sites of NoV Alpha, see FIG. 2). These positions were chosen because others had shown that these loops can tolerate large polypeptide insertions without disrupting capsid assembly. These modified capsid proteins can be referred to as enveloped FHV (eFHV) and enveloped NoV (eNoV). Codon optimized genes encoding the FHV and NoV Alpha proteins were synthesized by GeneArt (ThermoFisher), and cloned into a CMV and CAG-based mammalian expression vector (pCMV, DNASU ID: EvNO00601609) by Gibson assembly. To create detectable controls that should not be released from cells, Myc epitope tags alone were introduced at the indicated positions (FIG. 3A). To create the quasi-enveloped constructs, Myc-p6-PH sequences were amplified by polymerase chain reactions (PCR) from EPN-25, and transferred into the indicated positions by Gibson assembly. Flexible 22GS linkers to allow more flexibility C-terminal to the PH domain were inserted. Constructs for eFHV, eNoV and the controls were all verified by sequencing, and they can be detected by western blotting using anti-Myc antibodies. A schematic of the final constructs is shown in FIG. 3A.

(120) The polypeptide sequences of the ESCRT-recruiting HIV-1 p6^{sup}.Gag domain and the membrane-binding 13 N-terminal residues from Lyn Kinase that induce myristoylation and palmitoylation of the protein were inserted into the N terminus of AAV VP2. To uncouple expression of VP2 from VP1 and VP3, start codons for VP1 and VP2 were modified in the plasmid pXR2 by site-directed mutagenesis (see FIG. 7). A polynucleotide string comprising the Lyn1-13-p6-Myc sequence that also included a flexible 22GS linker C-terminal to the Myc-tag was synthesized and cloned into the pXR2-based VP2 expression vector by Gibson assembly. The N terminus of VP2 was truncated by around the horn mutagenesis.

(121) 2. Mammalian Cell Culture

(122) All experiments were carried out in human embryonic kidney (HEK) 293T cells obtained from American Type Culture Collection (ATCC) and cultured at 37° C. and 5% CO₂ in D-MEM media (ThermoFisher) containing 10% FBS, penicillin (100 U/ml) and streptomycin (0.1 mg/ml). Cells were tested for *Mycoplasma* contamination every 3 months using the MycoAlert™ *Mycoplasma* Detection Kit (Lonza).

(123) 3. Viron Release Assays

(124) To assay release of the designed FHV and NoV constructs, 8×10⁵ 293T cells were seeded in 6-well-plates 24 h prior to transfection. Cells were transfected with 2 µg of plasmid DNA expressing the FHV and NoV constructs (FIG. 3B), or co-transfected with 1 µg of plasmids expressing the FHV or NoV constructs and 1 µg of either pEGFP-VPS4A(E228Q) or pEGFP-C1 (Clontech), using the polyethyleneimine method (PEI, Polysciences, 3 µl of PEI/µg DNA). The transfection media was replaced with 1.5 ml growth media 5 h later. Cells and culture supernatants were harvested 24 h post transfection. Released vesicles were collected from the culture supernatants by centrifugation through a 200 µl 20% sucrose cushion for 90 min at 21,000×g, 4° C., and denatured by adding 30 µl 1× Laemmli SDS-PAGE buffer and boiling for 5 min. Cells were lysed in 100 µl cold lysis buffer (50 mM Tris pH 7.4, 150 mM NaCl, 1% Triton X-100, protease inhibitors) and 100 µl 2× Laemmli buffer supplemented with 10% 2-mercaptoethanol (Sigma) and boiled for 10 min. 15 µl samples of the cellular lysates and released vesicles were separated by 8% SDS-PAGE, transferred onto PVDF membranes, and probed with antibodies against the Myc epitope. GAPDH protein was used as a loading control. Protein bands were visualized by probing the membrane with fluorescently labelled secondary antibodies (Li-Cor Biosciences), and scanning with an Odyssey Imager (Li-Cor Biosciences). Levels of expressed and released Myc-tagged proteins were quantified by western blot densitometry using ImageJ, and comparison to standard curves generated with known quantities of recombinant EPN-01* protein produced in *E. coli*. Release efficiencies are reported as the percentage of protein in pelleted supernatants vs. the sum of the protein in cells and pelleted supernatants.

(125) FIG. 3B shows that eFHV eAlpha and eNoV capsids are released (lanes 2-4 and 6-8) whereas control Alpha constructs are not (lanes 1 and 5). Thus, the designed polypeptide insertions are required to drive efficient release. Note, Alpha constructs are released in the presence of eAlpha constructs (lanes 2, 3, 6, and 7), indicating that the proteins form an assembly. FIG. 3C shows that NoV eAlpha capsids are released (lanes 2-4, and 6-8) whereas control Alpha constructs are not (lanes 1 and 5). Thus, the designed polypeptide insertions are required to drive efficient release. Note, Alpha constructs are released in the presence of eAlpha constructs (lanes 2, 3, 6, and 7), indicating that the proteins form an assembly. FIG. 4B shows that overexpression of a dominant negative (DN-) VPS4A enzyme, which inhibits the ESCRT pathway, completely blocks the release of eFHV and eNoV (lanes 4 and 8). Thus, vesicle release is ESCRT dependent, as designed. Note that some “release” of non-enveloped NOV was detected when DN-VPS4A was co-expressed (lane 5), but this could reflect cell toxicity resulting from DN-VPS4A overexpression.

(126) 4. Vesicle Purification

(127) Released vesicles used in biochemical and cryo-EM studies were purified from culture supernatants of 293T cells (T225 flask, 6×10⁶ cells seeded per flask) following transient

transfection with plasmids encoding eFHV and eNoV (30 µg/flask) using the PEI method. Transfected cells were incubated overnight and the media was replaced after 5 h with exosome production media (D-MEM supplemented with 10% FBS, depleted of contaminating extracellular particles by centrifugation overnight at 100,000×g at 4° C., and subsequently filtered through a 0.22 µm filter). Cells were grown for 48 h and extracellular vesicles were purified by a series of filtration and centrifugation steps. Briefly, cell debris was removed by centrifugation of the supernatant at 1,000×g for 5 min followed by filtration through a 0.22 µm filter (EMD Millipore). Vesicles were collected by centrifugation at 100,000×g in an SW32Ti (BeckmanCoulter) at 4° C. for 1 h. Pellets were resuspended in PBS and pooled in a single tube (SW41 rotor, BeckmanCoulter). PBS was added to fill the tube and vesicles were collected by centrifugation at 100,000×g at 4° C. for 1 h. Pellets were resuspended in 1 ml of PBS and concentrated by centrifugation at 100,000×g at 4° C. for 1 h in an OptimaMAX-E (BeckmanCoulter) benchtop ultracentrifuge using a TLS-55 rotor. FHV and NOV release levels were quantified by western blotting as described above.

(128) 5. Protease Protection Assays

(129) To test for membrane envelopment, purified vesicles were resuspended and incubated under three different conditions, 10 µl each: (i) untreated sample (FIG. 4A lane 1), (ii) sample+0.05 mg/ml trypsin (FIG. 4A lane 2), and (iii) sample+0.05 mg/ml trypsin+1% Triton X-100 (FIG. 4A lane 3). Samples were incubated for 30 min at 25° C. 1 mM PMSF was then added and the samples were incubated for 10 min at 25° C. to inactivate trypsin. Samples were denatured by boiling for 10 min in 4× Laemmli buffer supplemented with 5% 2-mercaptoethanol, and analyzed by SDS-PAGE/western blot using an anti-Myc antibody. FIG. 4A shows that both released eFHV and eNoV proteins were protected against trypsin digestion in the absence of detergent, but became susceptible to trypsin digestion in the presence of 1% Triton X-100.

(130) 6. Cryo-EM Tomographic Imaging of Vesicles

(131) To prepare samples for cryo-EM tomography, 3 µl of purified vesicles in PBS were mixed with 3 µl of BSA-coated gold fiducials (10 nm size, Electron Microscopy Sciences). 3.5 µl samples of the suspension were applied to glow-discharged R2/2 holey carbon coated EM grids (Quantifoil), within the environmental chamber of a Vitrobot Mark IV (FEI) maintained at 4° C., 80% relative humidity. Excess liquid was blotted from the grids for 1.5 s (blot force 20, 1 blot) with filter paper (Whatman), before plunge freezing in liquid ethane. Cryo-grids were placed in a Gatan 626 cryoholder (Gatan) and imaged in a 200 kV Tecnai TF20 microscope (FEI) equipped with a K2 summit direct electron detector (Gatan). Tilt series were recorded bidirectionally starting from 0° to ±~600 with a 30 step size at a magnification of 22,500× and a defocus of ~8 µm (total dose per specimen ~300 e-/Å²), using the low-dose mode in SerialEM. Tomograms were generated using the IMOD software package. Image stacks were aligned and binned by 4 within IMOD. Aligned image stacks were Fourier filtered (cutoff 0.25, sigma 0.08) and tomographic reconstructions were performed using the simultaneous reconstruction technique (SIRT). Noise reduction was performed with the non-linear anisotropic diffusion (NAD) method in IMOD, using a K value of 0.04 with 12 iterations. FIG. 5B shows a slice from a tomogram that contains an icosahedral structure of the right size for FHV capsid (arrows), inside a vesicle membrane (arrowheads).

(132) Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific embodiments of the method and compositions described herein. Such equivalents are intended to be encompassed by the following claims.

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Claims

1. A modified virus comprising: a) one or more capsids, wherein each of the one or more capsids comprises modified capsid proteins comprising a capsid forming protein, a membrane binding element and an endosomal sorting complex required for transport (ESCRT)-recruiting element, wherein the capsid forming protein is a capsid forming protein of a non-enveloped virus, wherein at least one of the membrane binding element and the ESCRT-recruiting element is heterologous to the capsid forming protein; and b) a membrane surrounding the one or more capsids, wherein the membrane does not comprise viral fusion proteins.
 2. The modified virus of claim 1, wherein the membrane binding element is a membrane binding element of an enveloped virus.
 3. The modified virus of claim 1, wherein the membrane binding element is membrane binding element of a non-enveloped virus.
 4. The modified virus of claim 1, wherein the membrane binding element is a non-viral membrane binding protein.
 5. The modified virus of claim 1, wherein the membrane binding element is a non-naturally occurring protein.
 6. The modified virus of claim 1, wherein the ESCRT-recruiting element is an ESCRT-recruiting element of an enveloped virus.
 7. The modified virus of claim 1, wherein the ESCRT-recruiting element is an ESCRT-recruiting element of a non-enveloped virus.
 8. The modified virus of claim 1, wherein the membrane binding element and the ESCRT-recruiting element are a membrane binding element and an ESCRT-recruiting element of the same virus.
 9. The modified virus of claim 1, wherein the membrane binding element is the membrane binding PH element from phospholipase C.
 10. The modified virus of claim 1, wherein the membrane binding element is the membrane binding myristoylated and palmitoylated element of Lyn kinase.
 11. The modified virus of claim 1, wherein the ESCRT-recruiting element is the p6.sup.Gag polypeptide from human immunodeficiency virus.
 12. The modified virus of claim 1, wherein the ESCRT-recruiting element comprises any of SEQ ID NOs: 22, 23, 46-51, 154, 161-189.
 13. The modified virus of claim 1, further comprising a desired cargo.
 14. A method of treating a subject comprising administering a composition comprising the modified virus of claim 1 to a subject in need thereof.
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